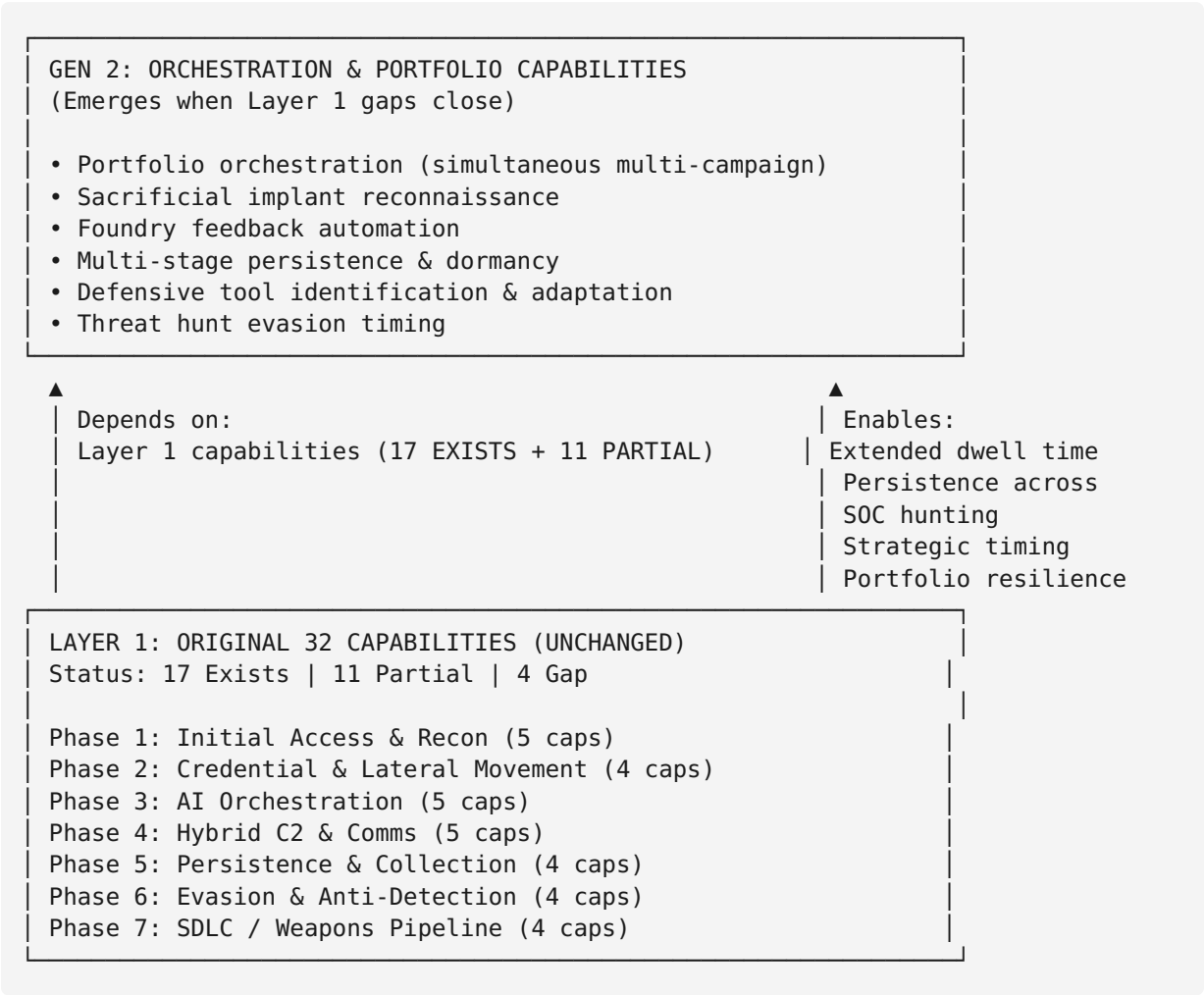


Gen 2 Capabilities: Layered Architecture for the Hydra Tracker

Architecture Overview



Gen 2 Capability Groups

Group G2-1: Portfolio Orchestration

Capability G2-1.1: Multi-Implant Simultaneous Management

Definition: Orchestrate 10+ implants across multiple hosts simultaneously, each with different risk profile, C2 channel, and objective.

Dependencies:

- ✓ Layer 1: ao-2 (Real-time tool selection)
- ✓ Layer 1: c2-1 (C2 via legitimate services)
- ✓ Layer 1: c2-4 (Dynamic C2 rotation)
- ? Layer 1: ao-5 (Adaptive evasion) [PARTIAL - sufficient for basic switching]

Current Status: PARTIAL → Tier 1-2 actors likely doing this manually

Timeline to Exists: 6 months (automation is the gap)

What it is:

- Implant A: Sacrificial probe on Host 1 (expect detection)
- Implant B: Persistent backdoor on Host 2 (dormant)
- Implant C: Lateral move implant on Host 3 (coordinated timing)
- Implant D-J: Other strategic implants

Each implant:

- Has independent C2 channel
- Reports to central orchestration
- Has different evasion rules
- Executes on independent schedule

Central orchestrator:

- Tracks status of all implants
- Decides which to activate/deactivate
- Coordinates lateral movement sequence
- Manages shared credentials/findings

Weighting factors:

- Complexity: High (10+ implants, independent C2)
- Detection risk: Medium (sacrificial ones caught, others hidden)
- ROI: Very high (resilience against detection)

Capability G2-1.2: Risk-Tiered Deployment Strategy

Definition: Classify and manage implants by risk profile (Pawns/Rooks vs Knights vs King) with accept-loss vs preserve strategies.

Dependencies:

- ✓ Layer 1: ao-1 (Recursive problem-solving) [determines if worth deploying]
- ✓ Layer 1: ao-3 (Environment value assessment) [determines deployment strategy]
- ? Layer 1: sd-3 (Field-to-foundry) [feedback from pawns → foundry response]

Current Status: PARTIAL → Tier 1-2 actors doing this via human judgment

Timeline to Exists: 12 months (requires ao-3 maturity)

What it is:

- PAWNS: Sacrificial implants, 100% loss rate expected
Risk tolerance: Infinite (accept any loss)
Count: 5-10 per campaign
Purpose: Intelligence gathering, SOC probing
- KNIGHTS/BISHOPS: Persistent backdoors, 0% loss expected
Risk tolerance: Minimal (preserve at all costs)
Count: 3-10 per campaign
Purpose: Long-term access, reactivation capability

- KING/QUEEN: Strategic objective implant, deployed only when ready
Risk tolerance: Medium (high risk, but only when success likely)
Count: 1 per campaign
Purpose: Crown jewel exfiltration or supply chain attack

Weighting factors:

- Requires ao-3 to determine which org deserves which tier
- Requires sd-3 to feedback pawn intelligence to foundry
- High strategic value: Protects king/queen with pawn/knight cover

Group G2-2: Sacrificial Intelligence Gathering

Capability G2-2.1: Intentional Loss for SOC Reconnaissance

Definition: Deploy expendable implants specifically to trigger detection, measure SOC response, and inform persistence strategy.

Dependencies:

- ✓ Layer 1: ao-2 (Real-time tool selection) [select different C2 per probe]
- ✓ Layer 1: c2-1, c2-2, c2-3 (Multiple C2 techniques)
- ✓ Layer 1: ev-1 (Polymorphic code) [vary each sacrificial implant]
- ? Layer 1: ao-5 (Adaptive evasion) [observe what triggers detection]

Current Status: NEAR-EXISTS → Malware can do this today, needs coordination
Timeline to Exists: 3-6 months (tactical skill, not gap)

What it is:

- Deploy 5-10 different implant variants
- Each on a separate, low-value host
- Each uses different C2 channel
- Each exhibits different behavior:
 - * Variant 1: Obvious HTTP beacon
 - * Variant 2: File creation
 - * Variant 3: Registry write
 - * Variant 4: Memory-resident only
 - * Variant 5: Legitimate API call

Measure:

- Which variant detected first? (indicates detection type)
- Detection latency? (2min vs 2h vs 2 days = Tier 1 vs Tier 3 vs Tier 5)
- Response action? (blocked vs isolated vs left running = SOC maturity)
- EDR behavior? (which processes checked, which files scanned)

Output:

- SOC classification (Tier 1-5)
- Optimal C2 channel to avoid detection
- Evasion tactics that work against THIS SOC
- Persistence locations that are safe

Cost:

- 5-10 sacrificial implants (accepted loss)
- 2-4 hours of probing

Benefit:

- Perfect intelligence about SOC response
- Guides entire persistence strategy
- Informs what persistent backdoors will avoid

Weighting factors:

- Cost: Very low (implants are expected to be caught)
- Benefit: Very high (intelligence is perfect, not inferred)
- Risk to main campaign: Zero (separate implants)
- Strategic value: Extremely high (enables everything else)

Capability G2-2.2: EDR/AV Specific Evasion Tuning

Definition: Based on identified EDR (CrowdStrike vs Sentinel vs McAfee), tune persistence strategy and C2 to maximize evasion.

Dependencies:

- ✓ Layer 1: ev-1 (Polymorphic code)
- ✓ Layer 1: ev-3 (AI real-time evasion adaptation) [enhanced version]
- ✓ From G2-2.1: Knowledge of which EDR is installed

Current Status: PARTIAL → Detection is possible, tuning is new

Timeline to Exists: 6-9 months (requires EDR-specific behavioral models)

What it is:

If CrowdStrike Falcon detected:

- Avoid: Obvious process anomalies
- Avoid: Registry modifications
- Use: Memory-resident implants only
- Use: Legitimate process injection
- Use: Slow C2 (6h+ beacon intervals)
- Reason: CrowdStrike monitors behavioral deviation

If Microsoft Defender detected:

- Avoid: Known malware signatures
- Use: Polymorphic code mutations
- Use: Legitimate service abuse
- Use: Standard C2 (behavioral less tuned)
- Reason: Defender monitors file signatures more than behavior

If McAfee Legacy detected:

- Avoid: Disk-based artifacts
- Use: Memory-resident, file-less
- Use: Any C2 (file monitoring only)
- Use: Aggressive operation (detection latency is days)
- Reason: McAfee has minimal behavioral detection

Group G2-3: Foundry Feedback Automation (sd-3 Enhancement)

Capability G2-3.1: Autonomous Field Intelligence Reporting

Definition: Field implant autonomously decides what intelligence to report and structures it for foundry consumption without human instruction.

Dependencies:

- ✓ Layer 1: ao-2 (Tool/exploit selection) [informs what to report]
- x Layer 1: sd-3 (Currently RED - this is the automation gap)

Current Status: PARTIAL → Field gathering works, reporting is manual

Timeline to Exists: 6-12 months (automation + structuring)

What it needs to report:

1. Environment profile
 - OS versions, patch levels
 - Installed software (dev tools, databases, etc)
 - Network topology (flat vs segmented)
2. Security posture
 - EDR installed? (type + version)
 - AV installed? (type + version)
 - Firewall rules (inferred from C2 blocking)
 - SIEM presence (inferred from response latency)
3. Access profile
 - Current privilege level
 - Available credentials
 - Lateral move opportunities
 - Crown jewel locations
4. Strategic opportunity assessment
 - Is this worth 6-month campaign?
 - Is this 2-week quick-hit org?
 - What's the optimal objective?

Autonomous decision:

- Implant decides "I have gathered enough intel, time to retreat"
- Implant structures report in machine-readable format
- Implant sends to foundry via secure channel
- Implant waits for foundry response

Weighting factors:

- Difficulty: High (requires decision logic + report structuring)
- Value: Extreme (enables foundry to build custom exploits)
- Timeline: 6-12 months (Tier 1-2 actors)

Capability G2-3.2: Autonomous Custom Exploit Generation

Definition: Foundry (AI or humans) automatically generates custom exploits based on field intelligence within hours, not weeks.

Dependencies:

- ✗ Layer 1: sd-3 (Currently RED - this is the bottleneck)
- ✓ Layer 1: Exploit library (existing vulnerabilities mapped to orgs)

Current Status: PARTIAL → Humans can do this in 2-4 weeks

Timeline to Exists: 6-12 months (automation of human process)

Current timeline (manual):

T+0h: Field reports findings
T+4h: Analyst reads report
T+8h: Team brainstorm what to build
T+24h: Exploit dev team codes
T+72h: QA tests
T+96h: Deploy to field
Total: ~4 days

Desired timeline (autonomous):

T+0h: Field reports findings (structured data)
T+15min: AI analyzes environment

T+30min: AI searches exploit library for matches
T+1h: AI generates custom payload (evasion-tuned)
T+1.5h: AI validates via simulation
T+2h: AI sends to field
Total: ~2 hours

What it generates:

1. Custom exploit for unpatched vuln in THIS org
2. Evasion payload tuned for THIS EDR
3. C2 protocol tuned for THIS org's traffic patterns
4. Lateral move sequence for THIS network topology
5. Persistence strategy for THIS environment

All in <2 hours, based on field intelligence

Weighting factors:

- Difficulty: Very high (requires ML + exploit generation)
- Value: Extreme (20x faster custom exploits)
- Timeline: 12-18 months (significant research required)
- Impact: Compresses threat timeline dramatically

Group G2-4: Multi-Stage Persistence & Dormancy

Capability G2-4.1: Strategic Retreat & Foundry Synchronization

Definition: Initial implant deliberately retreats, plants backdoor, waits for optimized implants from foundry before resuming operation.

Dependencies:

- ✓ Layer 1: c2-1 (Legitimate protocol C2) [for backdoor]
- ✓ Layer 1: pc-1 (Multi-mechanism persistence)
- ? Layer 1: ao-3 (Know when retreat is optimal)

Current Status: PARTIAL → Retreat and waiting possible, timing optimization is new
Timeline to Exists: 6-9 months (timing logic + coordination)

What it is:

Stage 1: Aggressive recon (2-4h)

- Deploy noisy implant
- Gather intelligence
- Trigger detection expected

Stage 2: Rapid retreat

- Kill noisy implant
- Plant simple backdoor in memory
- Go completely dormant
- Send intelligence report to foundry

Stage 3: Foundry response (2-6h with automation)

- Foundry analyzes intelligence
- Builds custom exploits + evasion
- Sends optimized implants

Stage 4: Deployment of optimized implants

- Backdoor receives custom implants
- Deploys lateral move exploit
- Establishes underground C2

- Plants persistent backdoors

Stage 5: Strategic dormancy

- All active operations cease
- Minimal C2 (1x per day)
- Wait for optimal exfiltration window

Decision logic:

- When to retreat? (after intel gathered + foundry contacted)
- When to deploy optimized? (when foundry has custom exploits)
- When to activate objective? (when threat hunt team leaves + security team distracted)

Weighting factors:

- Coordination complexity: High (multi-stage sequencing)
- Timing risk: Medium (wrong timing = detection)
- Persistence value: Extreme (backdoors survive both stages)

Capability G2-4.2: Dormancy Scheduling & Threat Hunt Evasion

Definition: Persistent backdoors remain dormant on precise schedule, evading threat hunts by being inactive during active hunting, reactivating after hunt team goes home.

Dependencies:

- ✓ Layer 1: ao-5 (Adaptive evasion) [detect hunt, go dormant]
- ? Layer 1: c2-5 (Legitimate user mimicry) [appear inactive]

Current Status: PARTIAL → Can go dormant, hunt detection is weak

Timeline to Exists: 9-12 months (threat hunt timing detection)

What it is:

Dormancy schedule:

- Default: C2 beacon 1x per 24 hours (minimal)
- When SOC hunting detected: Stop all beacons (complete silence)
- Duration: Until hunt team goes home (inference from activity patterns)

Hunt detection (from G2-2 sacrificial probes):

- EDR telemetry spike (more processes being inspected)
- Firewall blocking (C2 channels blocked)
- Process scanning (forensic tools like autoruns)
- Unusual SIEM correlation (rule engine more active)

Inference of hunt end:

- No more forensic tool execution (hunt team stopped working)
- SIEM rule activity decreased
- EDR telemetry returned to baseline
- Time-based (usually 4-8 hours for hunt cycle)

Reactivation:

- Once hunt appears over, minimal C2 resumes
- Only if hunt didn't find persistent implants
- Else: Stay dormant indefinitely

Weighting factors:

- Difficulty: Medium (hunt detection + timing)
- Value: Extreme (backdoors survive active hunts)
- Realism: High (actual threat actor behavior)

Sankey Weighting Model

Concept

Instead of binary status (Exists/Partial/Gap), use **weighted probability flows**:

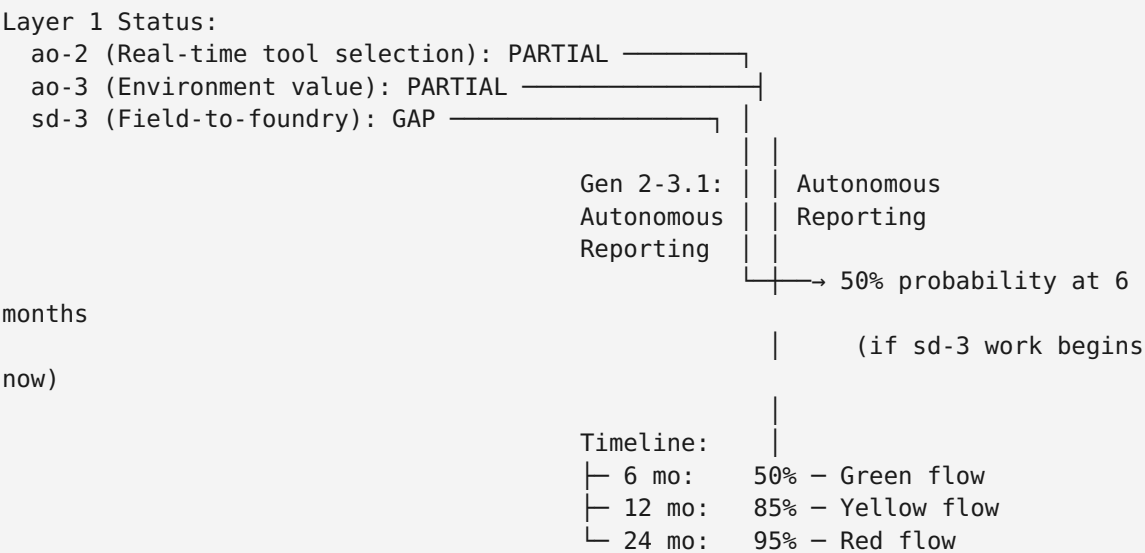
Layer 1 Capability Status → Gen 2 Capability Dependency → Gen 2 Capability Realization

Thickness of flow = Weighted probability capability will mature to this state in time period

Color = Timeline (6 months = green, 12 months = yellow, 24 months = red)

Arrows = Dependency relationships

Example Sankey Flow: G2-3.1 (Autonomous Reporting)



Multi-Year Threat Trajectory (Sankey Timeline)

TODAY (June 2026)

Layer 1: 17 EXISTS | 11 PARTIAL | 4 GAP
Gen 2: Mostly RESEARCH (0% exists)

↓

6 MONTHS (Dec 2026)

Layer 1: 19 EXISTS | 11 PARTIAL | 2 GAP (ao-3 closes)
Gen 2: Early adoption (5-15% of Tier 1 actors)

- ✓ G2-1.1 (Portfolio orchestration): 20% probability
- ✓ G2-2.1 (Sacrificial probes): 60% probability
- ✓ G2-3.1 (Autonomous reporting): 40% probability
- G2-3.2 (Custom exploit gen): 10% probability
- G2-4.1 (Strategic retreat): 30% probability

↓

12 MONTHS (June 2027)

Layer 1: 20 EXISTS | 10 PARTIAL | 2 GAP (c2-5 partially closes)

Gen 2: Tier 1-2 implementation (30-50% of advanced actors)

- ✓ G2-1.1 (Portfolio orchestration): 60% probability
- ✓ G2-2.1 (Sacrificial probes): 85% probability
- ✓ G2-3.1 (Autonomous reporting): 70% probability
- ✓ G2-3.2 (Custom exploit gen): 50% probability
- ✓ G2-4.1 (Strategic retreat): 70% probability
- ✓ G2-4.2 (Dormancy scheduling): 40% probability

↓

24 MONTHS (June 2028)

Layer 1: 22 EXISTS | 8 PARTIAL | 2 GAP (most capabilities mature)

Gen 2: Widespread Tier 1-2 implementation (70%+ of advanced actors)

- ✓ All G2 capabilities: 75%+ probability

THREAT LEVEL: Extreme (multi-stage, multi-vector, persistent)

How to Use This in the Tracker

1. Keep Layer 1 (32 capabilities) Unchanged

The original whitepaper framework remains the authoritative baseline. Status updates continue as-is.

2. Add Gen 2 as Separate Layer

New capability cards, tracked separately:

```
{
  "id": "g2-2-1",
  "name": "Intentional Loss for SOC Reconnaissance",
  "group": "G2-2: Sacrificial Intelligence Gathering",
  "status": "near_exists",
  "probability_exists": {
    "6_months": 0.60,
    "12_months": 0.85,
    "24_months": 0.95
  },
  "dependencies": [
    "ao-2",    // Real-time tool selection
    "c2-1",    // C2 via legitimate services
    "ev-1"     // Polymorphic code
  ],
  "enables": [
    "g2-1-1",  // Multi-implant orchestration
    "g2-2-2",  // EDR-specific evasion
    "g2-3-2"   // Custom exploit generation
  ],
  "threat_actor_capability": {
    "tier_1_2": "High - likely doing manually",
  }
}
```

```

    "tier_3": "Medium - planning phase",
    "tier_4_5": "Low - not yet"
  },
  "detection_evasion_gain": "Perfect SOC intelligence, enables 20-40% additional dwell time",
  "timeline_to_exists": "3-6 months (tactical skill, not research gap)"
}

```

3. Sankey Visualization

X-axis: Timeline (Today → 6mo → 12mo → 24mo) Y-axis: Threat actor tier (Tier 5 → Tier 1)
 Thickness: Weighted probability Color: Timeline (green=6mo, yellow=12mo, red=24mo)

Flows show: - Layer 1 capabilities flowing into Gen 2 capabilities - Gen 2 capabilities flowing into threat outcome (dwell time, persistence, resilience)

4. Live Intel Updates

Continue tracking threat intel as-is, but tag which Gen 2 capability it advances:

```

{
  "id": "intel-sacrifice-probes-observed",
  "date": "2026-06-15",
  "title": "APT35 Observed Deploying Multiple Test Implants",
  "source": "GTIG",
  "summary": "Chinese APT group observed deploying 7 different implant variants...",
  "g2_capability_impact": "g2-2-1 (Intentional Loss for Recon)",
  "status_change": "g2-2-1 advances from NEAR_EXISTS to PARTIAL",
  "probability_update": {
    "previous": 0.60,
    "new": 0.75,
    "timeline_compressed": "From 6mo to 4mo"
  }
}

```

Key Benefits of This Architecture

Benefit	Explanation
Baseline stability	Original 32 capabilities unchanged, authoritative
Layered sophistication	Gen 2 shows what emerges when Layer 1 matures
Probabilistic realism	Weighted probabilities, not binary status
Dependency mapping	Shows which Layer 1 gaps block which Gen 2 capabilities
Timeline compression	Sankey shows when Gen 2 becomes real threat (6-24mo)
Intel integration	New findings update Gen 2 probabilities in real-time
Gap prioritization	Shows which Layer 1 gaps are CRITICAL for Gen 2
Tier differentiation	Shows Tier 1 vs Tier 5 actor capability gaps

Critical Insight from This Architecture

The Framework's Real Threat is Not Layer 1 Capabilities.

It's **Gen 2 capabilities emerging from Layer 1 maturity.**

The Hydra becomes truly dangerous when: 1. Layer 1 capabilities are mostly Exists/Partial (done ✓) 2. sd-3 becomes automated (6-12 months away) 3. ao-3 matures (12-18 months away) 4. Gen 2 portfolio strategy reaches Tier 1-2 actors (12-24 months away)

The 12-24 month timeline is the real threat window.

After that, multi-stage portfolios with sacrificial probing become standard Tier 1-2 offensive tradecraft.